UrbanSense: An AI-Powered Edge Computing Framework for Motor Vehicle Analytics and Road Defect Telemetry

***Abstract*—The rapid expansion of smart city infrastructure necessitates robust, real-time telemetry systems capable of processing high-resolution urban data at the edge. This paper presents UrbanSense, a comprehensive computer vision pipeline engineered for dual-purpose municipal deployment: Motor Vehicle Analytics (MVA) and Road Defect Detection. The MVA module integrates YOLO-based tracking, ByteTrack algorithms, and Optical Character Recognition (OCR) to autonomously identify traffic infractions such as helmet absence, wrong-way driving, and pedestrian crossing violations. Concurrently, the road surface monitoring module leverages the Unified Road Defect Dataset to train lightweight vision models capable of categorising pavement degradation. By processing video feeds locally on Tesla T4-equivalent edge accelerators, UrbanSense minimises latency and centralises evidence logging into an accessible SQLite architecture, providing a scalable solution for urban mobility and asset management.**

***Index Terms*—Computer Vision, Edge Computing, Smart Cities, Traffic Analytics, Road Surface Monitoring, YOLO**

I. Introduction

Modern municipal governance relies heavily on the continuous collection and analysis of urban telemetry. Traditional methods of traffic enforcement and road surface inspection are labour-intensive, prone to human error, and struggle to scale with the increasing demands of urban mobility. The advent of edge computing and lightweight deep learning architectures has provided a paradigm shift, enabling high-fidelity video analytics to be processed directly at the source.

The UrbanSense platform addresses these challenges through a unified pipeline designed for deployment on edge devices. By isolating inference tasks from cloud-dependent architectures, the system guarantees real-time responsiveness and data privacy. This documentation outlines the design, training, and deployment methodology of the UrbanSense platform, detailing its two primary operational modules: the Motor Vehicle Analytics (MVA) pipeline and the Road Defect Telemetry system.

II. Comparative Literature Review

The integration of artificial intelligence into smart city infrastructure has been the subject of extensive recent research, particularly concerning edge deployment and mobile telemetry.

Zhang et al. [1] proposed a highly effective edge-computing paradigm for real-time traffic and road surface monitoring. By leveraging public transit fleets equipped with sensor arrays, their framework decentralised computational loads, significantly reducing the bandwidth required for municipal data transmission. Their methodology highlights the efficiency of mobile edge nodes but faces challenges regarding computational resource constraints on standard transit vehicles.

Similarly, Sharma and Gupta [2] conducted a comprehensive review of AI-powered mobile urban crowdsourcing. Their research emphasised the viability of distributing smart city telemetry across public and private mobile devices, creating a dynamic, self-updating map of urban infrastructure. While crowdsourcing provides vast spatial coverage, the authors noted that maintaining data quality and algorithmic normalisation across heterogeneous edge devices remains a significant hurdle.

In contrast to decentralised crowdsourcing, the UrbanSense architecture focuses on deploying fixed, highly optimised vision models to municipal edge nodes (such as CCTV infrastructure), ensuring consistent data normalisation and reliable enforcement auditing.

III. System Architecture: Motor Vehicle Analytics (MVA)

The MVA module is designed to enforce traffic regulations autonomously. The pipeline ingests live video feeds, tracks unique entities, evaluates spatial infractions, and logs actionable evidence.

A. Object Detection and Tracking

The detection engine is built upon the YOLO architecture, specifically optimised for small object recovery such as motorcycle helmets and licence plates. The model was trained using a custom 5-class dataset categorising Vehicles, Riders, Helmets, No-Helmets, and Number Plates.

To maintain entity identities across fast-moving frames, the pipeline employs ByteTrack. The tracker configuration is heavily customised for urban environments, utilising a highly forgiving parameter set (`track_buffer: 30`, `fuse_score: True`) to prevent identity loss during occlusion events common in dense traffic.

B. Violation Heuristics

The system enforces three primary heuristics:

- **Inverse-Helmet Detection:** Rather than relying solely on the presence of a helmet, the algorithm calculates the Intersection over Union (IoU) between a dynamically mapped anatomical head region (the upper 35% of the rider's bounding box) and both 'Helmet' and 'No-Helmet' classes. A violation is triggered if a bare head is explicitly detected or if no verified helmet overlaps the region.

- **Pedestrian Stop-Line Breaches:** A configurable polygon outlines the zebra crossing zone. The system flags an infraction if the central coordinate of a tracked vehicle intersects this spatial threshold.
- **Wrong-Way Motion Profiling:** A trajectory vector is calculated over a rolling 15-frame temporal window. If a vehicle exhibits a negative displacement vector against the legal flow of traffic, a wrong-way violation is logged.

C. Evidence Persistence and OCR

Upon detecting an infraction, the system extracts a bounding box crop of the offending vehicle and rider. It sequentially scans adjacent regions for a valid licence plate. If located, EasyOCR extracts the alphanumeric string, which is then mapped against an internal Regional Transport Office (RTO) registry. The evidence, including cropped image paths, timestamp, and entity details, is committed to an SQLite database. In testing, the pipeline successfully retrieved and logged 79 distinct violation records from a standard 90-second urban traffic sample.

IV. System Architecture: Road Defect Telemetry

Building upon our team's original work regarding the deployment of lightweight computer vision models on edge devices for municipal asset management [3], the second phase of the UrbanSense pipeline automates the categorisation of pavement degradation.

A. Dataset Configuration

The system utilises the `Unified_Road_Defect_Dataset`, an extensive repository of diverse urban surface conditions. The data was extracted and formatted to conform to the YOLO directory structure, yielding a robust corpus of 25,677 training images and 4,509 validation images.

The dataset is mapped to four primary infrastructural defects:

1) **Class 0:** Longitudinal Cracks (D00)
2) **Class 1:** Transverse Cracks (D10)
3) **Class 2:** Alligator Cracks (D20)
4) **Class 3:** Potholes (D40)

B. Pipeline Execution

During operation, the edge accelerator normalises input frames to an 800×800 resolution to preserve the fine textural details required for crack delineation. Bounding box coordinates are processed and overlaid in real-time, allowing municipal authorities to generate dynamic heatmaps of road decay, prioritising maintenance based on defect severity and frequency.

V. Conclusion

The UrbanSense platform demonstrates the efficacy of edge-computed computer vision in modernising municipal infrastructure. By integrating advanced tracking algorithms with highly targeted violation heuristics, the MVA module provides a scalable, automated solution for traffic enforcement. Concurrently, the road defect telemetry module offers a proactive approach to asset management. Together, these systems reduce the computational overhead typically associated with cloud analytics while maximising the utility of existing urban camera networks. Future iterations will focus on optimising the inference speed for hardware-constrained micro-edge devices, further democratising smart city telemetry.

References

[1] J. Zhang, et al., "Edge-Computing-Based Real-Time Traffic and Road Surface Monitoring Using Public Transit Fleets," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 4110-4125, 2023.

[2] M. Sharma and R. Gupta, "AI-Powered Mobile Urban Crowdsourcing: A Review of Smart City Infrastructure Telemetry," *Journal of Urban Mobility and Automation*, vol. 12, pp. 88-104, 2025.

[3] A. Kumar and S. Das, "Deploying Lightweight Computer Vision Models on Edge Devices for Municipal Asset Management," *International Conference on Smart Automation*, vol. 1420, no. 2, p. 052021, 2024.